Assignment1:

AIM: Write a program to count the total number of tokens and to identify and classify
them as keywords, identifiers, operators, constants and separators.

C Program:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

const char *keywords[] = {
    "int", "float", "char", "if", "else", "while", "for", "return",
    "break", "continue", "void", "switch", "case", "default", "do",
    "double", "long", "short", "unsigned", "signed", "const", "static"
};

int isKeyword(char *word) {
    int n = sizeof(keywords) / sizeof(keywords[0]);
    for (int i = 0; i < n; i++) {
        if (strcmp(word, keywords[i]) == 0)
            return 1;
    }
    return 0;
}

int main() {
    char input[1000], token[100];
    int i = 0, j = 0;

    int keywordCount = 0, identifierCount = 0;
    int operatorCount = 0, constantCount = 0, separatorCount = 0;
```

```
printf("Enter a line of code:\n");
fgets(input, sizeof(input), stdin);

while (input[i] != '\0') {


    if (isspace(input[i])) {
        i++;
        continue;
    }

      if (isalpha(input[i]) || input[i] == '_') {
        j = 0;
        while (isalnum(input[i]) || input[i] == '_') {
            token[j++] = input[i++];
        }
        token[j] = '\0';

        if (isKeyword(token)) {
            printf("%s -> Keyword\n", token);
            keywordCount++;
        } else {
            printf("%s -> Identifier\n", token);
            identifierCount++;
        }
    }

    else if (isdigit(input[i])) {
        j = 0;
        while (isdigit(input[i])) {
            token[j++] = input[i++];
        }
        token[j] = '\0';
```

```
            printf("%s -> Constant\n", token);
            constantCount++;
        }


        else if (strchr("+-*/=%<>!", input[i])) {
            printf("%c -> Operator\n", input[i]);
            operatorCount++;
            i++;
        }

              else if (strchr(",;(){}[]", input[i])) {
            printf("%c -> Separator\n", input[i]);
            separatorCount++;
            i++;
        }

        else {
            i++; // ignore unknown characters
        }
    }

    printf("Keywords: %d\n", keywordCount);
    printf("Identifiers: %d\n", identifierCount);
    printf("Operators: %d\n", operatorCount);
    printf("Constants: %d\n", constantCount);
    printf("Separators: %d\n", separatorCount);

    return 0;
}
```

Example Input:

int a = 10;

int -> Keyword
a -> Identifier
= -> Operator
10 -> Constant
; -> Separator

Output:

Keywords: 1
Identifiers: 1
Operators: 1
Constants: 1
Separators: 1